School of Computer Science Engineering and Information Systems

BCSE302L- Database Systems

COURSE PROJECT GUIDELINES – DATABASE SYSTEMS

1. Objective

The course project is intended to enable students to apply the concepts learned in the Database Systems laboratory to develop a real-world database application. The project should demonstrate how SQL and PL/SQL components work together to support real-world transactions and database operations.

2. Project Theme

Students shall develop a database application for a suitable real-world domain, such as:

Banking | Hospital | Library | E-Commerce | Hotel | Inventory | Railway/Bus Reservation | Online Examination | Employee/Payroll | Vehicle Rental | College Management, etc.

The selected application should involve multiple entities, relationships and meaningful database transactions.

3. Minimum Project Requirements

The project shall include the following:

- 6–8 related tables with appropriate Primary Keys and Foreign Keys.
- Appropriate constraints such as NOT NULL, UNIQUE, CHECK and DEFAULT.
- At least 10 meaningful records per table.
- An ER Diagram and corresponding relational schema.
- At least 15 meaningful SQL queries covering data retrieval, joins, aggregation, subqueries and other relevant SQL operations.
- At least 2 Procedures and 2 Functions to implement meaningful business operations.
- At least 1 Cursor for processing multiple records.
- At least 2 Triggers for automatic database actions.
- Demonstration of INSERT, UPDATE and DELETE operations.
- Database Connectivity with Java, Python, or any other suitable programming language to demonstrate interactive communication between the application and the database.

4. Interactive Database Demonstration

The project should demonstrate the flow:

User Operation → SQL/PL/SQL → Database Processing → Database Update/Retrieval → Result

For each major transaction, students should be able to explain:

- What operation is being performed?
- Which table(s) are involved?
- Which SQL/PL/SQL component is executed?
- How is the database affected?
- What result is generated?

The project should not be a collection of independent SQL/PLSQL programs. Every component must have a clear purpose in the proposed application.

5. Project Report

The report shall contain:

1. Project Title and Team/Student Details
2. Introduction and Problem Statement
3. Functional Requirements
4. ER Diagram
5. Relational Schema
6. Table Creation and Constraints
7. SQL Queries and Outputs
8. Procedures, Functions, Cursor and Triggers
9. Real-World Transaction Demonstrations
10. Conclusion

REVIEW I – Problem Analysis & Database Design

Evaluation Component	Evaluation Criteria	Marks
Problem Definition & Scope	Clear problem statement, objectives, users and scope of the proposed system	2
Requirement Analysis	Identification of major entities, attributes and functional requirements	2
ER Diagram	Correct entities, attributes, relationships, cardinality and keys	3
Relational Schema	Correct conversion of ER model into relational tables with PK/FK relationships	2
Viva / Understanding	Ability to explain the proposed system and justify design decisions	1
Total		**10**

REVIEW II – Database Implementation

Evaluation Component	Evaluation Criteria	Marks
DDL & Constraints	Creation of tables with appropriate PK, FK and other constraints	2
DML & Data Population	Meaningful insertion, modification and deletion of data	1
SQL Queries	Meaningful queries using joins, aggregation, subqueries, grouping, etc.	2
Procedures & Functions	Appropriate implementation of business operations using Procedures and Functions	2
Cursor & Triggers	Meaningful use of Cursor and Triggers in the proposed application	2
Viva / Technical Understanding	Ability to explain SQL/PLSQL implementation and execution	1
Total		**10**

REVIEW III – Integrated System & Final Demonstration

Evaluation Component	Evaluation Criteria	Marks
Integrated Database Operation	Database components work together to support the proposed application	2
Real-World Transactions	Demonstration of at least 5 meaningful transactions/operations	3
SQL/PLSQL Interaction	Appropriate use of queries, procedures, functions, cursors and triggers during transactions	2
Output & Reports	Correct and meaningful results/reports generated from the database	1
Project Report & Documentation	Completeness, clarity and technical documentation	1
Viva / Project Understanding	Ability to explain the complete database workflow and justify design choices	1
Total		**10**